The Journal of Open Source Software

# ebtelplusplus: Efficient Hydrodynamic Modeling of Coronal Loops

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**DOI:** 10.xxxxxx/draft

**Software**

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**Editor:**

**Submitted:** 30 October 2025
**Published:** unpublished

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## Summary

Understanding the response of the plasma in the solar corona, the outermost layer of the Sun's atmosphere, to heating is of considerable importance in understanding flares and the heating of the quiescent corona. This requires detailed numerical modeling of the coupled system of the coronal plasma and the solar magnetic field[1]. While solving the full set of three-dimensional *magnetohydrodynamic* (MHD) equations is feasible for small regions of the corona, the needed computational resources and physical complexity of such models means they are not always amenable to simple or even correct interpretation. Field-aligned hydrodynamic models (e.g. HYDRAD Bradshaw & Mason, 2003) exploit the fact that the magnetic pressure in the corona is much greater than the gas pressure such that the corona can be considered as a series of mini atmospheres, or *coronal loops*, where the plasma responds hydrodynamically to the heating and traces out the complex coronal magnetic field. As such, the explicit dependence on the magnetic field can be neglected and the relevant hydrodynamic equations can be reduced to a single-dimension in space: the coordinate along the coronal loop. However, the large range of spatial and temporal scales necessary to resolve this system, in particular the severe time step limitations imposed by thermal conduction, mean that even field-aligned models are computationally expensive enough to make large parameter explorations prohibitive. The enthalpy-based thermal evolution of loops (EBTEL) model (Cargill et al., 2012; Klimchuk et al., 2008) was originally developed in order to provide a simple and efficient way to study the coronal plasma response to time-dependent plasma heating. EBTEL accomplishes this by computing spatial integrals of the aforementioned field-aligned hydrodynamic equations. Comparisons to spatially-averaged results from field-aligned hydrodynamic models show very good agreement (Cargill et al., 2012). Because of its relative simplicity and computational efficiency, EBTEL has been widely used since its initial development (e.g. Qiu et al., 2012; Ugarte-Urra & Warren, 2014) with Klimchuk et al. (2008) and Cargill et al. (2012) having nearly 400 citations combined according to the Astrophysics Data System.

## Statement of Need

The key to EBTEL lies in its treatment of enthalpy between the corona and the transition region (TR), the narrow layer of the solar atmosphere that connects the corona with the

[1]This also applies to the coronae of F, G, K, and M dwarf stars.

denser chromosphere below. This enables EBTEL to split the coronal loop into two regions that match across the corona/TR boundary, leading to a set of coupled, non-linear ordinary differential equations that model the spatially-averaged, time-dependent behavior of the relevant thermodynamic quantities. EBTEL equates an enthalpy flux with an imbalance between the heat flux out of the corona and the energy lost due to radiation in the TR. If TR radiation cannot balance the downward heat flux, this drives an upflow of material into the corona and if the TR is radiating away more energy than the coronal heat flux can supply this drives a downflow. This approximation is valid for bulk velocities below the local sound speed (Klimchuk et al., 2008).

The original software implementation of the EBTEL model of Klimchuk et al. (2008) was in the proprietary Interactive Data Language (IDL). Subsequent improvements to the gravitational stratification and radiative losses by Cargill et al. (2012) gave better agreement with field-aligned hydrodynamic models[2]. Barnes et al. (2016) modified the EBTEL model to relax the single-fluid assumption and treat electrons and ions separately, allowing for differential heating between the two species and implemented these modifications in C++. Cargill et al. (2021) later extended EBTEL to include effects due to cross-sectional area expansion and Reep et al. (2024) added the ability to vary the abundance model for the radiative losses as a function of time.

`ebtelplusplus` unifies all of the aforementioned features into a single set of equations and C++ and Python software implementation. In particular, `ebtelplusplus` solves the following equations for the spatially-averaged electron pressure ($p_e$), ion pressure ($p_i$), and number density ($n$) of a semi-circular coronal loop of half-length $L$,

$$
\begin{aligned}
\frac{1}{\gamma-1}\frac{dp_e}{dt} &= Q_e + \frac{\psi_c}{L_*}\left(1+\frac{A_{TR}\psi_{TR}}{A_c\psi_c}\right) - \frac{R_c}{L_*}\left(1+c_1\frac{A_{TR}}{A_c}\right), \\
\frac{1}{\gamma-1}\frac{dp_i}{dt} &= Q_i - \frac{\psi_c}{L_*}\left(1+\frac{A_{TR}\psi_{TR}}{A_c\psi_c}\right), \\
\frac{dn}{dt} &= -\frac{(\gamma-1)\xi c_2}{(\xi+1)\gamma c_3 k_B L_c T_e}\left(\frac{A_{TR}L_c}{A_c L_*}R_c\left(c_1-\frac{L_{TR}}{L_c}\right)+\frac{A_0}{A_c}(F_{e,0}+F_{i,0})\right),
\end{aligned}
$$

where $Q_{e,i}$ are the user-specified heating terms for the electrons and ions, $\psi_{c,TR}$ denote integrals of the electron-ion coupling terms over the TR and corona, respectively, $A_{c,TR,0}$ are the cross-sectional area averaged over the corona and TR and at the TR/corona boundary, respectively, $R_c$ is the energy lost to radiation in the corona, $c_1$ is the ratio of energy lost to radiation in the TR and corona, $\xi = T_e/T_i$ is the ratio between the electron and ion temperatures, $F_{e,0;i,0}$ are the conductive heat fluxes at the TR/corona boundary for the electrons and ions, $L_{c,TR}$ are the lengths of the corona and TR such that $L = L_c + L_{TR}$, and $L_* = L_c + (A_{TR}/A_c)L_{TR}$. The remaining terms are fixed constants. This set of equations is closed by an ideal gas law for the electrons and ions: $p_e = k_B n T_e, p_i = k_B n T_i$ and $n_e = n_i = n$ due to the assumption of a fully-ionized hydrogen plasma. These equations and their derivations are explained more fully in the aforementioned publications and the `ebtelplusplus` documentation.

`ebtelplusplus` solves the above equations using a Runge-Kutta Cash-Karp integration method (see section 16.2 of Press et al., 1992) and an (optional) adaptive time-stepping scheme to ensure the principal physical timescales are resolved at each phase of the loop evolution[3]. Where appropriate, all inputs and outputs are expressed as `astropy.units.Quantity` objects (Astropy Collaboration et al., 2022). Additionally, `ebtelplusplus` is very fast (a single run modeling $10^4$ seconds of simulation time takes only a few milliseconds) and nearly two orders of magnitude faster than the previous IDL implementation because it is implemented in C++

[2]This version is sometimes referred to as "EBTEL2".

[3]The Runge-Kutta Cash-Karp integrator is provided by the Boost Odeint library.

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and due to its use of an adaptive time-stepping scheme. Figure 1 shows example output from `ebtelplusplus` with different model parameters for the same time-dependent heating function.

`ebtelplusplus` is implemented in C++ for computational efficiency and is wrapped in Python using `pybind11` (Jakob et al., 2025) to enable easier installation and a user-friendly API. Precompiled binary wheels are built using `cibuildwheel` run on GitHub Actions[4] and distributed via PyPI at every release. These wheels are provided for all major operating systems and the versions of Python recommended by SPEC 0[5]. This alleviates the need for a user to compile the C++ code locally. `ebtelplusplus` is openly-developed on GitHub. Documentation, including an example gallery and a guide to contributing to the package, is hosted online on Read the Docs. `ebtelplusplus` also includes a comprehensive test suite built on the `pytest` testing framework that is run on GitHub Actions at each check-in. Test coverage is assessed using Codecov.

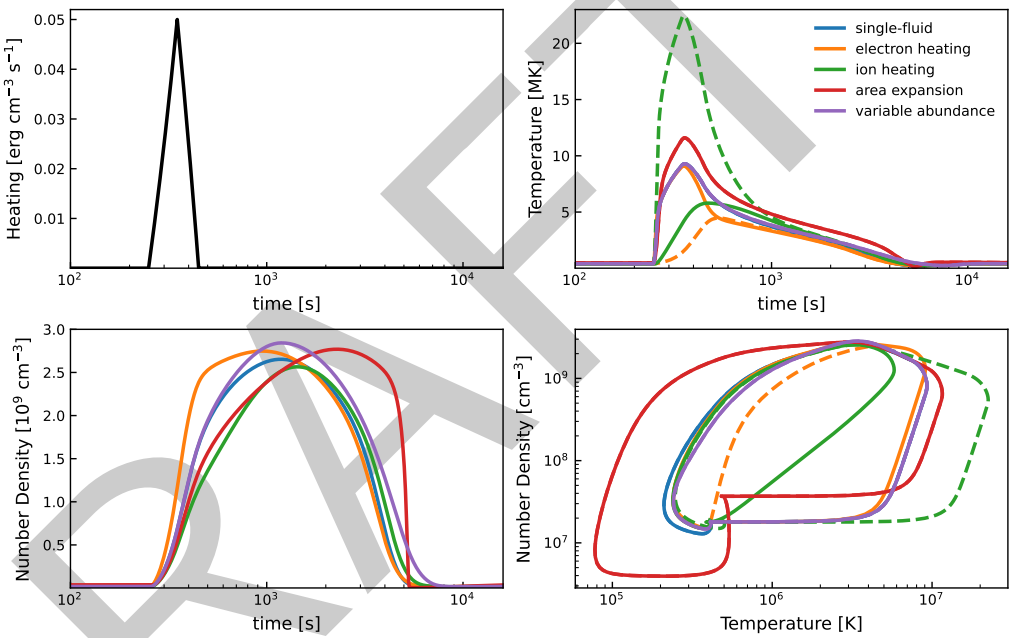


**Figure 1:** Temperature (top right), density (bottom left), and temperature-density phase space (bottom right) of a coronal loop with half-length $L = 40$ Mm for five different cases with the same heating input (top left panel). In the nominal case (blue), the electron and ion populations are kept in equilibrium, the cross-sectional area of the loop is constant, and the radiative losses are determined by a power-law function. If the electrons (solid) and ions (dashed) are allowed to evolve separately, heating only the electrons (orange) causes the ions to take about 250 s to fully equilibrate with the electrons while heating only the ions (green) causes the ions to become over three times hotter than the electrons due to the relative inefficiency of ion thermal conduction. Incorporating area expansion through the corona (red) leads to a higher peak temperature and a more delayed peak in the density while calculating the radiative losses using a time-varying abundance (purple) leads to a slightly higher peak density.

# Other Implementations

The aforementioned IDL implementation, which includes features described in Cargill et al. (2012) and Cargill et al. (2021), is referred to as `EBTEL-IDL` (Cargill et al., 2024). Rajhans et al. (2022) relaxed the assumption of subsonic flows in EBTEL such that the Mach numbers and velocities produced are in better agreement with field-aligned hydrodynamic

[4] `ebtelplusplus` uses the reusable GitHub Actions workflows provided by the OpenAstronomy project.

[5] Scientific Python Ecosystem Coordination (SPEC) 0 recommends a set of minimum supported dependencies for packages commonly used across the scientific Python ecosystem, including Python.

simulations for some heating scenarios. The IDL software implementation of this model is referred to as `EBTEL3-IDL`. As such, there are currently three separate though slightly different implementations of the EBTEL model. The table below summarizes the features included in each implementation.

| Feature | Citation | `EBTEL-IDL` | `EBTEL3-IDL` | `ebtelplusplus` |
|---|---|---|---|---|
| Decouple electrons and ions | Barnes et al. (2016) | no | no | yes |
| Adaptive time step | Barnes et al. (2016) | no | yes | yes |
| Area expansion | Cargill et al. (2021) | yes | no | yes |
| Supersonic flows | Rajhans et al. (2022) | no | yes | no |
| Time-variable abundances | Reep et al. (2024) | no | no | yes |

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